

Generalised Process Flow For Semiconductor Packaging



Curriculum is based on unit job processes!

internationally relevant, enabling professionals to work on semiconductor projects globally.

Are these courses already available in academia? Can a final-year engineering student start pursuing them?

Yes! As mentioned earlier, AICTE has already announced the revised curriculum. A final-year student can absolutely take these electives now.

So, who is working on building awareness amongst students about these opportunities and courses?

From 2024 until now, we (SEMI-ESSCI-ELCINA-IESA) have already covered the North, South, East, and West regions in our outreach. We have reached nearly 800 members of academia (faculty and students) and aim to cross 1000 by the end of July. At present, activity is slower due to exams and holidays, but workshops will resume shortly. Our goal is to ensure that the AICTE-approved electives are not merely notifications; they must be adopted by university boards of studies and implemented in classrooms. We are actively working to make this happen.

So, the AICTE approval is in place, but universities still need to adopt these electives, correct?

Correct. Some universities have

already begun. For example, we have conducted workshops and engagements at institutes like Manav Rachna, IIIT Bangalore, IIT Guwahati, and IIT Gandhinagar. The response has been very encouraging. Change is gradual; professors need to become comfortable with the subject, gain confidence in teaching it, and trust that they can do so without relying entirely on industry trainers. We are helping them along that journey.

How are faculty being trained? Are these workshops for that purpose?

Yes. The workshops we conducted across 50 to 60 institutes were primarily aimed at training both faculty and students. The aim was to inspire faculty and demonstrate that the curriculum is practical and self-executable. If needed, we are available to support them with content, clarification, and guidance.

If a professor wants support or content, who should they reach out to?

They can reach out to AICTE, and the request will be routed to the expert committee, which includes me and my colleagues. Alternatively, they can contact the Electronics Sector Skill Council of India (ESSCI) directly. ESSCI will forward the request to the skills committee, and we will respond. Professors who have already engaged

with us now understand the subject well and are familiar with what we are delivering. The support system is evolving rapidly.

We have been referring to 'engineers' throughout. Are these jobs primarily for engineering graduates, or are diploma holders eligible too?

That is an important question. These are technical roles that are ideal for technicians as well. In fact, engineering has, at times, become synonymous with design in our country, which is not helpful. Originally, we had created job roles such as HAZCHEM Safety Technician and Electrical Safety Technician. However, we changed the terminology from 'technician' to 'engineer' because many people are reluctant to accept the label of 'technician'. That said, **the opportunities remain wide open for diploma holders, and in many cases, they are more hands-on and can move ahead faster.**

So, diploma students may actually have a first-mover advantage in these roles?

Yes, absolutely. I have seen it firsthand. At NAMTECH in Gandhinagar, for instance, they have built excellent infrastructure for automation and robotics training. Some of their top-performing students come from diploma backgrounds, and they are already doing things many engineering graduates cannot.

There is a perception that fabs are heavily automated and will not generate many jobs. Is that true?

There is some truth to that. **Fabs are not going to employ millions. However, that does not mean the opportunity is insignificant. For instance, our conservative projection is 20,000-25,000 direct jobs in manufacturing and non-design roles.** MeitY estimates that the semiconductor industry will create 80,000 total jobs over the first five years. For example, Applied Materials